LIBRARY MANAGEMENT SYSTEM

Submitted By:

Harshita Negi (2410030521)

Saumya Kumar Singh (2410030511)

Palak Singh (2410030740)

Sarita Kumari (2410031612)

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Submitted To: Mr. Aditya Bangia

University: IILM University, Greater Noida

PROBLEM STATEMENT

A library has a large number of books, members, and daily transactions such as book issues and returns.

It is a time-consuming task to manage all these records manually.

The major problem to be addressed in this project is to store, manage and retrieve the data regarding:

- Books available in the library.
- Members of the library.
- Issuing and returning books.
- Due date and calculating fines.

The Library Management System will help to automate all these tasks using a database system. It will help librarians to manage digital records, reduce paperwork, remove duplicate data and increase accuracy and efficiency.

DATABASE SCHEMA

TABLE STRUCTURE:

Book Table:

```
CREATE TABLE Book (
    Book_ID INT PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(100),
    Publisher VARCHAR(100),
    Category VARCHAR(50)
);
```

Member Table:

```
CREATE TABLE Member (
    Member_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Contact_Number VARCHAR(15),
    Email VARCHAR(100),
    Membership_Date DATE
);
```

Librarian Table:

```
CREATE TABLE Librarian (
    Librarian_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Contact_Number VARCHAR(15),
    Email VARCHAR(100)
);
```

Issue Table:

```
CREATE TABLE Issue (
	Issue_ID INT PRIMARY KEY,
	Book_ID INT,
	Member_ID INT,
	Issue_Date DATE,
	Return_Date DATE,
	FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID),
	FOREIGN KEY (Member_ID) REFERENCES Member(Member_ID)

 );
```

Fine Table:

```
CREATE TABLE Fine (
	Fine_ID INT PRIMARY KEY,
	Issue_ID INT,
	Fine_Amount INT,
	Payment_Status VARCHAR(20),
	FOREIGN KEY (Issue_ID) REFERENCES Issue(Issue_ID)
 );
```

TABLE DATA

Book Table:

Result Grid | Filter Rows: | Edit: | Export/Import:

Book_ID	Title	Author	Publisher	Category
1	Database Systems	Korth	McGraw Hill	Education
2	Data Structures	Seymour Lipschutz	Schaum	Education
3	Physics Fundamentals	Resnick	Wiley	Science
4	Chemistry Basics	Morrison	Pearson	Science
5	English Grammar	Wren & Martin	S Chand	Language
6	Artificial Intelligence	Stuart Russell	Pearson	Technology
7	Operating Systems	Galvin	Wiley	Technology
8	Mathematics for Engineers	Grewal	Khanna Publishers	Education
9	Biology Today	Trueman	Trueman Publications	Science
NULL	NULL	NULL	NULL	NULL

Member Table:

Result Grid | Filter Rows: | Edit: | Export/Import:

Member_ID	Name	Contact_Number	Email	Membership_Date
101	Rahul Sharma	9876543210	rahul@gmail.com	2023-01-15
102	Priya Singh	9876501234	priya@gmail.com	2022-12-10
103	Amit Verma	9123456780	amit@gmail.com	2024-02-20
104	Sneha Gupta	9988776655	sneha@gmail.com	2023-05-25
105	Rohit Kumar	9871234567	rohit@gmail.com	2021-11-11
106	Karan Malhotra	9812345678	karan@gmail.com	2024-01-10
107	Anjali Mehta	9823456789	anjali@gmail.com	2023-07-18
108	Vikas Yadav	9834567890	vikas@gmail.com	2022-09-05
NULL	NULL	NULL	NULL	NULL

Librarian Table:

Result Grid | Filter Rows: | Edit:

Librarian_ID	Name	Contact_Number	Email
1	Anita Mehra	9998887776	anita@library.com
2	Rajesh Kumar	8887776665	rajesh@library.com
3	Neha Kapoor	9870001112	neha@library.com
NULL	NULL	NULL	NULL

Issue Table:

Result Grid | Filter Rows: | Edit:

Issue_ID	Book_ID	Member_ID	Issue_Date	Return_Date
1	1	101	2026-03-01	2026-03-10
2	3	102	2026-03-05	NULL
3	2	103	2026-03-07	2026-03-15
4	4	104	2026-03-10	NULL
5	6	101	2026-03-12	2026-03-20
6	7	106	2026-03-15	NULL
7	8	107	2026-03-16	2026-03-22
8	9	108	2026-03-18	NULL
NULL	NULL	NULL	NULL	NULL

Fine Table:

Fine_ID	Issue_ID	Fine_Amount	Payment_Status
1	1	50	Paid
2	2	30	Pending
3	3	20	Paid
4	4	40	Pending
5	6	60	Pending
6	7	10	Paid
NULL	NULL	NULL	NULL

SQL QUERIES

QUERY - 1 (Select with Condition)

Problem Statement:

Display all books in the Science category

SQL Query:

SELECT * FROM Book WHERE Category = 'Science';

Book_ID	Title	Author	Publisher	Category
3	Physics Fundamentals	Resnick	Wiley	Science
4	Chemistry Basics	Morrison	Pearson	Science
9	Biology Today	Trueman	Trueman Publications	Science

QUERY - 2 (Select with Condition)

Problem Statement:

Show members who joined after 2023

SQL Query:

SELECT Name , Membership_Date FROM Member WHERE Membership_date > '2023-01-01';

Name	Membership_Date
Rahul Sharma	2023-01-15
Amit Verma	2024-02-20
Sneha Gupta	2023-05-25
Karan Malhotra	2024-01-10
Anjali Mehta	2023-07-18

QUERY - 3 (JOIN)

Problem Statement:

Display which member borrowed which book

SQL Query:

SELECT Member.Name, Book.Title, Issue.Issue_Date
FROM Issue
JOIN Member ON Issue.Member_ID = Member.Member_ID
JOIN Book ON Issue.Book_ID = Book.Book_ID;

Name	Title	Issue_Date
Rahul Sharma	Database Systems	2026-03-01
Priya Singh	Physics Fundamentals	2026-03-05
Amit Verma	Data Structures	2026-03-07
Sneha Gupta	Chemistry Basics	2026-03-10
Rahul Sharma	Artificial Intelligence	2026-03-12
Karan Malhotra	Operating Systems	2026-03-15
Anjali Mehta	Mathematics for Engineers	2026-03-16
Vikas Yadav	Biology Today	2026-03-18

QUERY - 4 (JOIN + Condition)

Problem Statement:

Show all books that are not yet returned

SQL Query:

SELECT Book.Title, Member.Name
FROM Issue
JOIN Book ON Issue.Book_ID = Book.Book_ID
JOIN Member ON Issue.Member_ID = Member.Member_ID
WHERE Issue.Return_Date IS NULL;

Title	Name
Physics Fundamentals	Priya Singh
Chemistry Basics	Sneha Gupta
Operating Systems	Karan Malhotra
Biology Today	Vikas Yadav

QUERY - 5 (Count)

Problem Statement:

Count the total number of books

SQL Query:

SELECT COUNT(*) AS Total_Books FROM Book;

Total_Books
9

QUERY - 6 (GROUP BY)

Problem Statement:

Count number of books in each category

SQL Query:

SELECT Category, COUNT(*) AS Total FROM Book
GROUP BY Category;

Category	Total
Education	3
Science	3
Language	1
Technology	2

QUERY - 7 (SUM)

Problem Statement:

Calculate total fine collected

SQL Query:

SELECT SUM(Fine_Amount) AS Total_Fine FROM Fine
WHERE Payment_Status = 'Paid';

Total_Fine
80

QUERY - 8 (Subquery)

Problem Statement:

Find members who have borrowed books

SQL Query:

```
SELECT Name FROM Member
WHERE Member_ID IN (
   SELECT Member_ID FROM Issue
);
```

Name
Rahul Sharma
Priya Singh
Amit Verma
Sneha Gupta
Karan Malhotra
Anjali Mehta
Vikas Yadav

QUERY - 9 (UPDATE)

Problem Statement:

Update Return date of a book

SQL Query:

UPDATE Issue
SET Return_Date = '2026-03-25'
WHERE Issue_ID = 2;

Issue_ID	Book_ID	Member_ID	Issue_Date	Return_Date
1	1	101	2026-03-01	2026-03-10
2	3	102	2026-03-05	2026-03-25
3	2	103	2026-03-07	2026-03-15
4	4	104	2026-03-10	NULL
5	6	101	2026-03-12	2026-03-20
6	7	106	2026-03-15	NULL
7	8	107	2026-03-16	2026-03-22
8	9	108	2026-03-18	NULL

QUERY - 10 (AVERAGE)

Problem Statement:

Find the average fine amount

SQL Query:

SELECT AVG(Fine_Amount) AS Average_Fine
FROM Fine;

Average_Fine
35.0000

QUERY - 11 (MIN Aggregate)

Problem Statement:

Find the minimum fine amount

SQL Query:

SELECT MIN(Fine_Amount) AS Min_Fine
FROM Fine;

Min_Fine
10

QUERY - 12 (GROUP BY + HAVING)

Problem Statement:

Show categories having more than 2 books

SQL Query:

SELECT Category, COUNT(*) AS Total
FROM Book
GROUP BY Category
HAVING COUNT(*) > 2;

Category	Total
Education	3
Science	3

QUERY - 13 (JOIN + Aggregate)

Problem Statement:

Count how many books each member has borrowed

SQL Query:

SELECT Member.Name, COUNT(Issue.Book_ID) AS Books_Borrowed
FROM Issue
JOIN Member ON Issue.Member_ID = Member.Member_ID
GROUP BY Member.Name;

Name	Books_Borrowed
Rahul Sharma	2
Priya Singh	1
Amit Verma	1
Sneha Gupta	1
Karan Malhotra	1
Anjali Mehta	1
Vikas Yadav	1

QUERY - 14 LIKE (Pattern Matching)

Problem Statement:

Find members whose name starts with 'A'

SQL Query:

SELECT * FROM Member WHERE Name LIKE 'A%';

Member_ID	Name	Contact_Number	Email	Membership_Date
103	Amit Verma	9123456780	amit@gmail.com	2024-02-20
107	Anjali Mehta	9823456789	anjali@gmail.com	2023-07-18

INITIAL DATABASE SCHEMA

The following is the initial database schema we are using in our library management system.

Book Table:

```
CREATE TABLE Book (
    Book_ID INT PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(100),
    Publisher VARCHAR(100),
    Category VARCHAR(50)
);
```

Member Table:

```
CREATE TABLE Member (
    Member_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Contact_Number VARCHAR(15),
    Email VARCHAR(100),
    Membership_Date DATE
);
```

Librarian Table:

```
CREATE TABLE Librarian (
    Librarian_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Contact_Number VARCHAR(15),
    Email VARCHAR(100)
);
```

Issue Table:

```
CREATE TABLE Issue (
	Issue_ID INT PRIMARY KEY,
	Book_ID INT,
	Member_ID INT,
	Issue_Date DATE,
	Return_Date DATE,
	FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID),
	FOREIGN KEY (Member_ID) REFERENCES Member(Member_ID)

);
```

Fine Table:

```
CREATE TABLE Fine (
	Fine_ID INT PRIMARY KEY,
	Issue_ID INT,
	Fine_Amount INT,
	Payment_Status VARCHAR(20),
	FOREIGN KEY (Issue_ID) REFERENCES Issue(Issue_ID)
);
```

FUNCTIONAL DEPENDENCIES

Book Table:
Book_ID → Title, Author, Publisher, Category

Member Table:
Member_ID → Name, Contact_Number, Email, Membership_Date

Librarian Table:
Librarian_ID → Name, Contact_Number, Email

Issue Table:
Issue_ID → Book_ID, Member_ID, Issue_Date, Return_Date

Fine Table:
Fine_ID → Issue_ID, Fine_Amount, Payment_Status

These functional dependencies show us how each non-key attribute is dependent on the primary key of its respective table.

UNNORMALISED FORM (UNF)

Book_ID	Title	Author	Member_ ID	Member_ Name	Issue_Date	Return_Date	Fine_Amount
1	Database Systems	Korth	101	Rahul Sharma	2026-03-01	2026-03-10	50
3	Physics Fundamentals	Resnick	102	Priya Singh	2026-03-05	NULL	30
2	Data Structures	Seymour Lipschutz	103	Amit Verma	2026-03-07	2026-03-15	20
4	Chemistry Basics	Morrison	104	Sneha Gupta	2026-03-10	NULL	20
6	Artifical Intelligence	Stuart Russell	101	Rahul Sharma	2026-03-12	2026-03-20	40
7	Operating Systems	Galvin	106	Karan Malhotra	2026-03-15	NULL	60
8	Mathematics for Engineers	Grewal	107	Anjali Mehta	2026-03-16	2026-03-22	10
9	Biology Today	Trueman	108	Vikas Yadav	2026-03-18	NULL	NULL

The above table represents the Unnormalized Form (UNF), where all data is combined into a single table. This leads to redundancy and inefficient data management. Multiple attributes depend on different entities, making the structure complex and prone to inconsistencies. Therefore, normalization is required.

This table does not follow normalization rules and requires decomposition into smaller tables.

FIRST NORMAL FORM (1NF)

Original Table(UNF):

Book_ID	Title	Author	Member_ ID	Member_ Name	Issue_Date	Return_Date	Fine_Amount
1	Database Systems	Korth	101	Rahul Sharma	2026-03-01	2026-03-10	50
3	Physics Fundamentals	Resnick	102	Priya Singh	2026-03-05	NULL	30
2	Data Structures	Seymour Lipschutz	103	Amit Verma	2026-03-07	2026-03-15	20
4	Chemistry Basics	Morrison	104	Sneha Gupta	2026-03-10	NULL	20
6	Artifical Intelligence	Stuart Russell	101	Rahul Sharma	2026-03-12	2026-03-20	40
7	Operating Systems	Galvin	106	Karan Malhotra	2026-03-15	NULL	60
8	Mathematics for Engineers	Grewal	107	Anjali Mehta	2026-03-16	2026-03-22	10
9	Biology Today	Trueman	108	Vikas Yadav	2026-03-18	NULL	NULL

Functional Dependencies:

Book_ID → Title, Author
Member_ID → Member_Name
(Issue_Date depends on combination)

1NF Table:

Book_ID	Title	Author	Member_ ID	Member_ Name	Issue_Date	Return_Date	Fine_Amount
1	Database Systems	Korth	101	Rahul Sharma	2026-03-01	2026-03-10	50
3	Physics Fundamentals	Resnick	102	Priya Singh	2026-03-05	NULL	30
2	Data Structures	Seymour Lipschutz	103	Amit Verma	2026-03-07	2026-03-15	20
4	Chemistry Basics	Morrison	104	Sneha Gupta	2026-03-10	NULL	20
6	Artifical Intelligence	Stuart Russell	101	Rahul Sharma	2026-03-12	2026-03-20	40
7	Operating Systems	Galvin	106	Karan Malhotra	2026-03-15	NULL	60
8	Mathematics for Engineers	Grewal	107	Anjali Mehta	2026-03-16	2026-03-22	10
9	Biology Today	Trueman	108	Vikas Yadav	2026-03-18	NULL	NULL

The table is converted into 1NF by ensuring that all attributes contain atomic values and there are no repeating groups in the table.

SECOND NORMAL FORM (2NF)

Original Table(1NF):

Book_ID	Title	Author	Member_ ID	Member_ Name	Issue_Date	Return_Date	Fine_Amount
1	Database Systems	Korth	101	Rahul Sharma	2026-03-01	2026-03-10	50
3	Physics Fundamentals	Resnick	102	Priya Singh	2026-03-05	NULL	30
2	Data Structures	Seymour Lipschutz	103	Amit Verma	2026-03-07	2026-03-15	20
4	Chemistry Basics	Morrison	104	Sneha Gupta	2026-03-10	NULL	20
6	Artifical Intelligence	Stuart Russell	101	Rahul Sharma	2026-03-12	2026-03-20	40
7	Operating Systems	Galvin	106	Karan Malhotra	2026-03-15	NULL	60
8	Mathematics for Engineers	Grewal	107	Anjali Mehta	2026-03-16	2026-03-22	10
9	Biology Today	Trueman	108	Vikas Yadav	2026-03-18	NULL	NULL

Functional Dependencies:

Book_ID → Title, Author
Member_ID → Member_Name
(Book_ID, Member_ID, Issue_Date) → Return_Date, Fine_Amount

Book details only depend upon Book_ID and Member details depend upon Member_ID, this results in partial dependency.

Decomposition:

BOOK TABLE:

Book_ID	Title	Author	Publisher	Category
1	Database Systems	Korth	McGraw Hill	Education
2	Data Structures	Seymour Lipschutz	Schaum	Education
3	Physics Fundamentals	Resnick	Wiley	Science
4	Chemistry Basics	Morrison	Pearson	Science
5	English Grammar	Wren & Martin	S Chand	Language
6	Artificial Intelligence	Stuart Russell	Pearson	Technology
7	Operating Systems	Galvin	Wiley	Technology
8	Mathematics for Engineers	Grewal	Khanna Publications	Education
9	Biology Today	Trueman	Trueman Publications	Science

MEMBER TABLE:

Member_ID	Name	Contact_Number	Email	Membership_Date
101	Rahul Sharma	9876543210	rahul@gmail.com	2023-01-15
102	Priya Singh	9876501234	priya@gmail.com	2022-12-10
103	Amit Verma	9123456780	amit@gmail.com	2024-02-20
104	Sneha Gupta	9988776655	sneha@gmail.com	2023-05-25
105	Rohit Kumar	9871234567	rohit@gmail.com	2021-11-11
106	Karan Malhotra	9812345678	karan@gmail.com	2024-01-10
107	Anjali Mehta	9823456789	anjali@gmail.com	2023-07-18
108	Vikas Yadav	9834567890	vikas@gmail.com	2022-09-05

ISSUE TABLE:

Issue_ID	Book_ID	Member_ID	Issue_Date	Return_Date
1	1	101	2026-03-01	2026-03-10
2	3	102	2026-03-05	NULL
3	2	103	2026-03-07	2026-03-15
4	4	104	2026-03-10	NULL
5	6	101	2026-03-12	2026-03-20
6	7	106	2026-03-15	NULL
7	8	107	2026-03-16	2026-03-22
8	9	108	2026-03-18	NULL

The above tables now represent the Second Normal Form (2NF) as the existing partial dependencies have been removed by separating the book and member table into respective independent tables. Each attribute now fully depends on the primary key of its table.

THIRD NORMAL FORM (3NF)

Original Tables(2NF):

BOOK TABLE:

Book_ID	Title	Author	Publisher	Category
1	Database Systems	Korth	McGraw Hill	Education
2	Data Structures	Seymour Lipschutz	Schaum	Education
3	Physics Fundamentals	Resnick	Wiley	Science
4	Chemistry Basics	Morrison	Pearson	Science
5	English Grammar	Wren & Martin	S Chand	Language
6	Artificial Intelligence	Stuart Russell	Pearson	Technology
7	Operating Systems	Galvin	Wiley	Technology
8	Mathematics for Engineers	Grewal	Khanna Publications	Education
9	Biology Today	Trueman	Trueman Publications	Science

MEMBER TABLE:

Member_ID	Name	Contact_Number	Email	Membership_Date
101	Rahul Sharma	9876543210	rahul@gmail.com	2023-01-15
102	Priya Singh	9876501234	priya@gmail.com	2022-12-10
103	Amit Verma	9123456780	amit@gmail.com	2024-02-20
104	Sneha Gupta	9988776655	sneha@gmail.com	2023-05-25
105	Rohit Kumar	9871234567	rohit@gmail.com	2021-11-11
106	Karan Malhotra	9812345678	karan@gmail.com	2024-01-10
107	Anjali Mehta	9823456789	anjali@gmail.com	2023-07-18
108	Vikas Yadav	9834567890	vikas@gmail.com	2022-09-05

ISSUE TABLE:

Issue_ID	Book_ID	Member_ID	Issue_Date	Return_Date
1	1	101	2026-03-01	2026-03-10
2	3	102	2026-03-05	NULL
3	2	103	2026-03-07	2026-03-15
4	4	104	2026-03-10	NULL
5	6	101	2026-03-12	2026-03-20
6	7	106	2026-03-15	NULL
7	8	107	2026-03-16	2026-03-22
8	9	108	2026-03-18	NULL

Functional Dependencies:

Issue_ID → Fine_Amount
Fine_Amount depends on Issue_ID, not directly on primary entity
This is known as Transitive Dependency.

Decomposition:

FINE TABLE:

Fine_ID	Issue_ID	Fine_Amount	Payment_Status
1	1	50	Paid
2	2	30	Pending
3	3	20	Paid
4	4	40	Pending
5	6	60	Pending
6	7	10	Paid

LIBRARIAN TABLE:

Librarian_ID	Name	Contact_Number	Email
1	Anita Mehra	9998887776	anita@library.com
2	Rajesh Kumar	8887776665	rajesh@library.com
3	Neha Kapoor	9876500002	neha@library.com

The database is now in Third Normal Form (3NF). Transitive dependencies have been removed by creating separate tables for fine and librarian details. All attributes are now directly dependent on their respective primary keys, ensuring data consistency and eliminating redundancy.

FINAL NORMALIZED TABLES

BOOK TABLE:

Book_ID	Title	Author	Publisher	Category
1	Database Systems	Korth	McGraw Hill	Education
2	Data Structures	Seymour Lipschutz	Schaum	Education
3	Physics Fundamentals	Resnick	Wiley	Science
4	Chemistry Basics	Morrison	Pearson	Science
5	English Grammar	Wren & Martin	S Chand	Language
6	Artificial Intelligence	Stuart Russell	Pearson	Technology
7	Operating Systems	Galvin	Wiley	Technology
8	Mathematics for Engineers	Grewal	Khanna Publications	Education
9	Biology Today	Trueman	Trueman Publications	Science

MEMBER TABLE:

Member_ID	Name	Contact_Number	Email	Membership_Date
101	Rahul Sharma	9876543210	rahul@gmail.com	2023-01-15
102	Priya Singh	9876501234	priya@gmail.com	2022-12-10
103	Amit Verma	9123456780	amit@gmail.com	2024-02-20
104	Sneha Gupta	9988776655	sneha@gmail.com	2023-05-25
105	Rohit Kumar	9871234567	rohit@gmail.com	2021-11-11
106	Karan Malhotra	9812345678	karan@gmail.com	2024-01-10
107	Anjali Mehta	9823456789	anjali@gmail.com	2023-07-18
108	Vikas Yadav	9834567890	vikas@gmail.com	2022-09-05

LIBRARIAN TABLE:

Librarian_ID	Name	Contact_Number	Email
1	Anita Mehra	9998887776	anita@library.com
2	Rajesh Kumar	8887776665	rajesh@library.com
3	Neha Kapoor	9876500002	neha@library.com

ISSUE TABLE:

Issue_ID	Book_ID	Member_ID	Issue_Date	Return_Date
1	1	101	2026-03-01	2026-03-10
2	3	102	2026-03-05	NULL
3	2	103	2026-03-07	2026-03-15
4	4	104	2026-03-10	NULL
5	6	101	2026-03-12	2026-03-20
6	7	106	2026-03-15	NULL
7	8	107	2026-03-16	2026-03-22
8	9	108	2026-03-18	NULL

FINE TABLE:

Fine_ID	Issue_ID	Fine_Amount	Payment_Status
1	1	50	Paid
2	2	30	Pending
3	3	20	Paid
4	4	40	Pending
5	6	60	Pending
6	7	10	Paid

CONCLUSION

The Library Management System developed in this project demonstrates how a relational database can efficiently manage and organize library operations such as book records, member details, issue/return transactions, and fine management. By implementing various SQL queries, meaningful insights can be extracted from the database quickly and accurately.

The process of normalization from Unnormalized Form (UNF) to Third Normal Form (3NF) helped in reducing data redundancy and eliminating anomalies, thereby improving data consistency and integrity. The final database structure is well-organized, efficient, and easy to maintain.

Overall, this project highlights the importance of proper database design and shows how DBMS concepts can be applied to solve real-world problems effectively.